MATH 464. HW#9. Lazizbek.

Discussing the results:

LP Primal resut:

(1) $\min z = \text{None}$

$x = \text{None}$

The problem is unbounded $\Rightarrow$

(2) $z^* = -\infty \Rightarrow$

(3) • I can make $z = c^T x$ as small as possible & stay <u>feasible</u>, then by Theorem 4.1 (Weak duality) theorem says, if a feasible point p exists, then $p^T b \leq c^T x = -\infty$! $\Rightarrow$

"min problem"

It is impossible to have p. s.t

(4) $p^T b \leq -\infty$. $\Rightarrow$

That is given Primal LP is unbundend & if a feasible point p exists, then $p^T b \leq -\infty$, impossible happens, therefore there's no feasible point $p \Rightarrow$

(5) Dual program is <u>infeasible</u> as in the result from the software result for Dual LP!